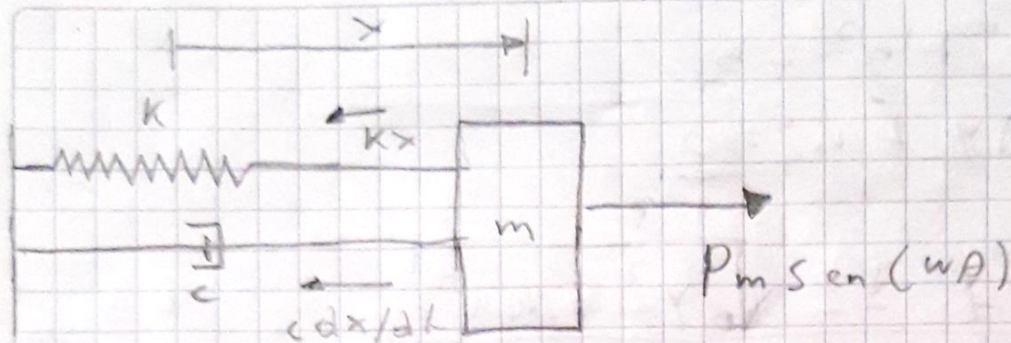


Ejercicio Entregable 3 Masa-Resorte

DD MM AA



$$m \ddot{x} + c \dot{x} + Kx = 0$$

x : desplazamiento

c : constante de amortiguamiento

K : constante de elasticidad

$$\dot{x} = v$$

$$\dot{v} = \ddot{x} = a = -\frac{c}{m} \dot{x} - \frac{K}{m} x$$

$$= -\frac{c}{m} v - \frac{K}{m} x$$

$$= \begin{pmatrix} \dot{x} \\ v \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ -\frac{K}{m} & -\frac{c}{m} \end{pmatrix}}_A \begin{pmatrix} x \\ v \end{pmatrix}$$

Valores Propios

$$\begin{pmatrix} \dot{x} \\ v \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\omega^2 & -\gamma \end{pmatrix} \begin{pmatrix} x \\ v \end{pmatrix}$$

$$\begin{pmatrix} \lambda - 1 \\ \omega^2 \lambda + \gamma \end{pmatrix} = 0$$

$$\gamma = \frac{c}{3K}$$

$$\omega = \sqrt{\frac{3K}{m}}$$

$$\omega^2 = \frac{3K}{m}$$

$$\lambda^2 + \gamma\lambda + \omega^2 = 0$$

$$\lambda^2 + \gamma\lambda + \omega^2 = 0$$

$$\lambda = \frac{-\gamma \pm \sqrt{\gamma^2 - 4\omega^2}}{2}$$

Para $\gamma^2 - 4\omega^2 < 0$ (sist. subamortiguado)

Se obtienen dos raíces complejas

$$-(4\omega^2 - \gamma^2) > 0$$

$$\lambda = \frac{-\gamma \pm \sqrt{(4\omega^2 - \gamma^2)(-1)}}{2}$$

$$\lambda = \frac{-\gamma \pm i\sqrt{4\omega^2 - \gamma^2}}{2}$$

$$\lambda_{1,2} = \left(-\frac{\gamma}{2} \right) \pm \frac{i\sqrt{4\omega^2 - \gamma^2}}{2}$$

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^{-\frac{\gamma}{2}t} \left(\cos\left(\frac{\sqrt{4\omega^2 - \gamma^2}}{2}t\right) + \text{sen}\left(\frac{\sqrt{4\omega^2 - \gamma^2}}{2}t\right) \right) \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

$$x(t) = e^{-\frac{\gamma}{2}t} \left(c_1 \cos\left(\frac{\sqrt{4\omega^2 - \gamma^2}}{2}t\right) + c_2 \text{sen}\left(\frac{\sqrt{4\omega^2 - \gamma^2}}{2}t\right) \right)$$